

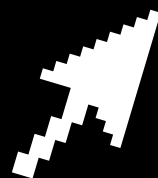
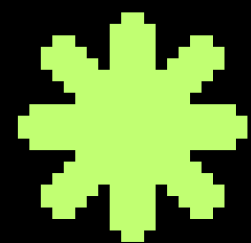


ANURA

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ELEVATOR PITCH

A small frog fights its way up the food chain in a 2D roguelite platformer set in a dangerous swamp. Players collect mosquitoes as persistent currency, build strategic card decks across runs, and defeat increasingly challenging predator bosses. Every card burns on use, choose wisely.

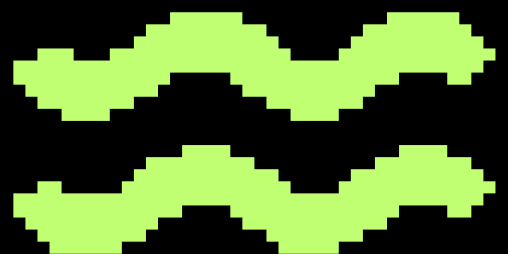


Run

Wow!



SMALL CREATURE,
BIG WORLD,
SMART SURVIVAL.



GAME OBJECTIVE

WHAT IS ANURA?

A 2D roguelite where you control a small frog surviving in a dangerous swamp.

- Navigate platforming sections
- Collect mosquitoes (currency)
- Face bosses with unique patterns
- Build strategic decks

El contraste entre exploración tranquila y combate intenso crea ritmo emocional.

La progresión está basada en aprendizaje estratégico, no solo reflejos o movimientos aleatorios.

MAIN GOAL

Survive and climb the food chain by defeating all bosses in a single run.

Each run:

1. Platforms → Boss 1
2. Platforms → Boss 2 (Final)

Death = progress

Buy cards → Try again stronger



CORE MECHANICS

MOVEMENT

A/D → Horizontal

W → Aim up

Space → Jump

Dash → (ONLY WITH CARD)

Glide → (ONLY WITH CARD)

Attack → I

TONGUE

- Primary weapon
- Collects mosquitoes
- Melee attack
- Hitbox system

CARDS

- 3 active slots
- Burn-and-replace
- Permanent effects (until burned)
- Strategy over Reflexes

Mosquitoes

- Never lost on death
- Persist across runs
- Spent on new cards
- Fair and permanent progression

BOSS SYSTEM (FSM)

Finite State Machine:

- Idle → Attack → Recovery → Phase 2
- Learnable patterns, not random
- Clear vulnerability windows
- Skill over Luck

ENEMY TYPES



mosquitoes

- Aggressive (least aggressive)
- Collectible with tongue
- In game currency



spiders

- Aggressive
- Patrol platforms
- Active obstacles



snake
(boss 1)

- Simple, predictable pattern
- Long recovery windows
- Teaches combat fundamentals
- HP: medium, Damage: medium



hawk
(boss 2)

- Fast, aggressive attacks
- Short vulnerability windows
- Phase 2 at 25% HP
- Final boss

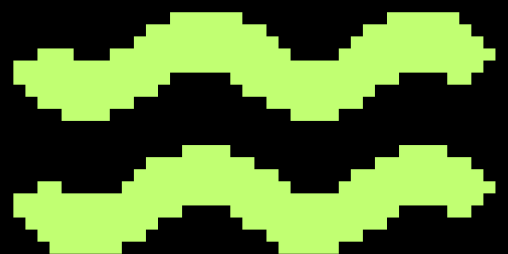
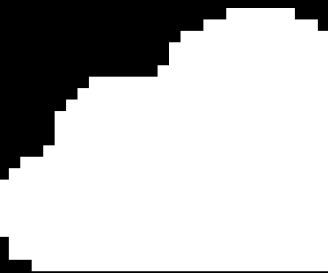


ROGUELITE ELEMENTS



1. Run
2. Defeat/Death
3. Reward (Mosquitoes)
4. Permanent Upgrade (Cards)
5. New Run Stronger

**"FAILURE IS
PROGRESS"**



TCG ELEMENTS

Strategic Card System

MOVEMENT

Mobility and traversal
abilities

Iron Hindlegs
Dragonfly Hop
Glide Membrane
Bubble Dash
Rocket Frog

COMBAT

Offensive and defensive
abilities

Chameleon Veil
Fire Kiss
Thunder Tongue
Toad Shockwave
Venom Lash

UTILITY

Support and resource
abilities

Lucky Pond
Metamorphosis
Spiked Whip
Tadpole Heart
Thorn Skin

Implemented TCG Mechanics:

Pre-run deck building, 3 limited slots, card
combinations benefit from each other, strategic
decisions, cooldowns and effects



VISUAL STYLE



Anura uses a pixel-art aesthetic with a "Cute but Deadly" design philosophy.

Swamp Surface

Calm, humid, cozy, slightly tense, natural and alive

Dense Swamp

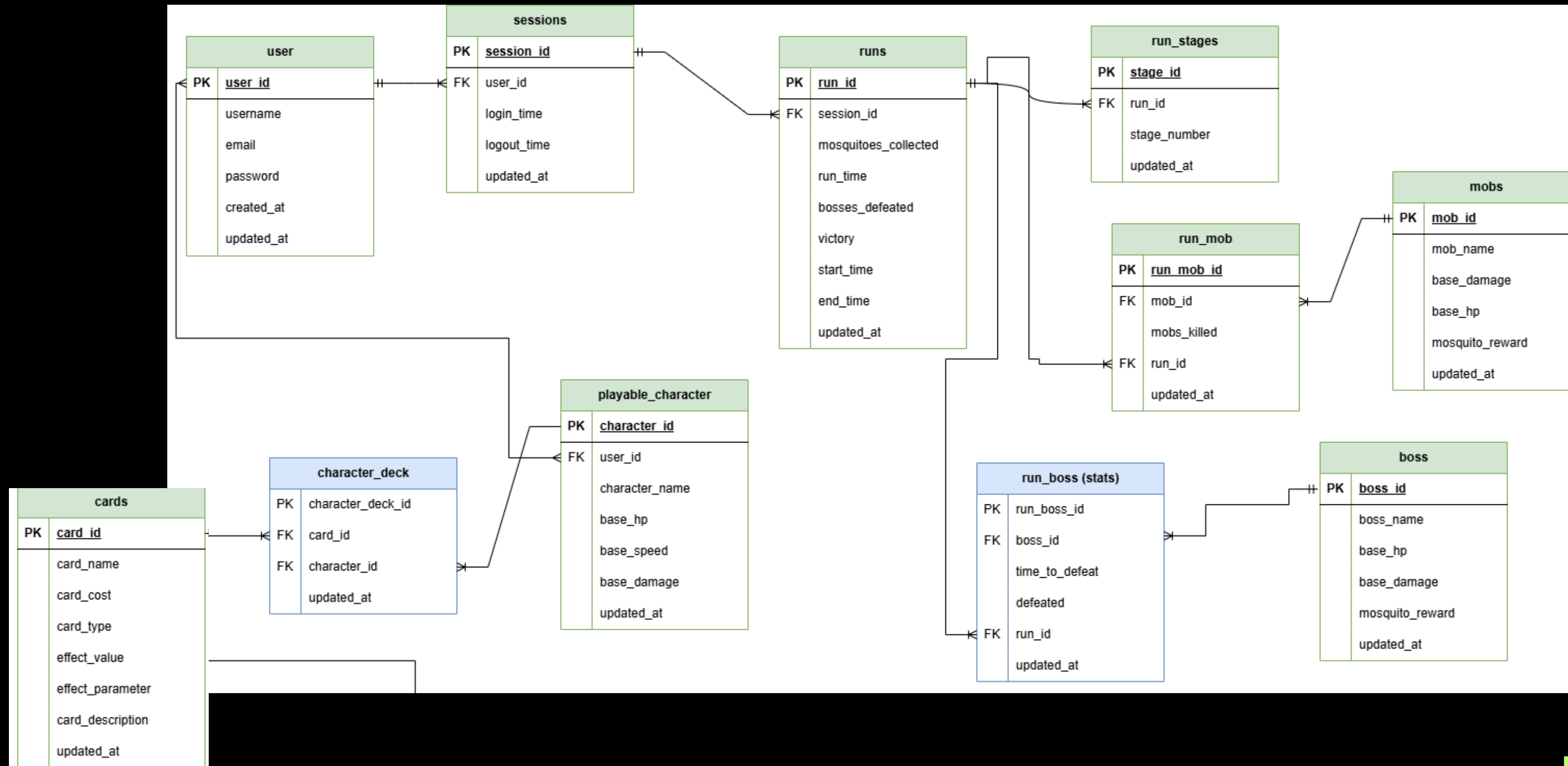
Darker, more enclosed, slightly oppressive, more dangerous, less visually open

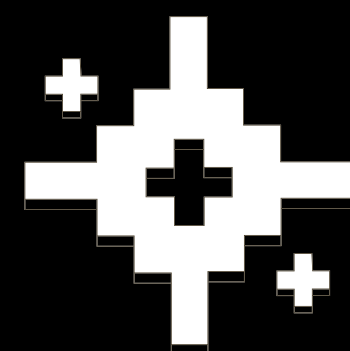
Predator Arena

Tense, focused, quiet before combat, isolated



DATABASE

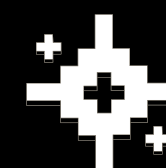




PART 2

GROUP

REFLECTION



WHY WE CHOSE THIS STRATEGY

WHAT WAS REQUIRED

- Build a roguelite
- Include TCG card mechanics
- Use a database

WHAT WE DECIDED

- Burn and replace card mechanic
- Persistent mosquito currency
- Platformer

We created a roguelite TCG platformer to combine action and strategy in one game. Each run feels different, and players build their playstyle through cards. The card selection screen we implemented lets players choose their deck, and the burn-and-replace system adds strategy by allowing them to replace unwanted cards.

CHALLENGES WE FACED

DATA PERSISTENCE + API

Problem: cards and mosquitoes had to persist across runs, cards burn on activation but survive death if unused, implementation was a struggle at first.

Solution: use of views for mysql

MySQL CONFIGURATION

Problem: 2 of the teammates kept having issues with the MySQL configuration, leading to errors, delays with the backend, etc.

Solution: use of mysql workbench community

DIFFERENT SCHEDULES & AVAILABILITY

Problem: the 3 of us had different schedules throughout the day, specially in the afternoon, which made organization difficult at first but we managed to adapt and find time for team work.

Solution: Microsoft Teams calls on weekends, weekdays, etc.

BACKEND

Problem: starting to work on the backend got us behind at first, we struggled a lot on understanding it, but after some classes and other resources we got the hang of it.

Solution: we kept asking our teacher's for help and even watched videos to learn more.

WHAT WE LEARNED

DATABASE:

- Data integrity
- CRUD Operations
- Views, Triggers, etc.
- Connection to backend

GAME DEVELOPMENT:

- Role of Physics in games
- FSM
- Videogame concepts

WEB DESIGN:

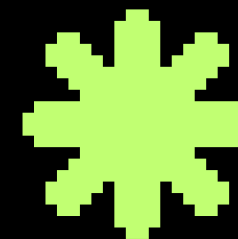
- How to create websites
- Use of css in HTML
- Bootstrap & JQuery

SOFTWARE DEVELOPMENT:

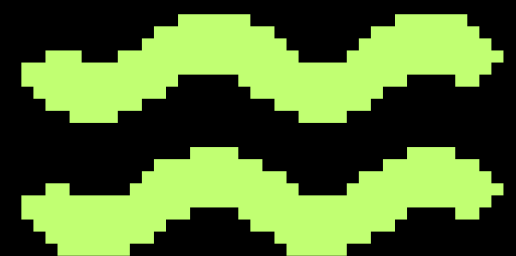
- Github
- SCRUM
- Github Projects
- TEAM WORK !!



This project was our introduction to full-stack development. We started with the basics like HTML and CSS, then worked our way up to building a complete system. We created a custom game engine in JavaScript for the frontend, built APIs using Express for the backend, used MySQL for the database, and connected everything together. Each part helped us learn something new.



FULL STACK DEVELOPMENT



TEAM REFLECTIONS

RENATA

Over these 10 weeks, I learned a lot and improved on my skills. At the beginning, I felt confused and didn't know many things, but now I feel happy with what I can do and that I helped build Anura. I also learned more about teamwork and realized how important it is for everything we do.

CARLOS

I learned how important organization and planning are in a project like this. Keeping everything structured was challenging at first, but over time we improved in managing tasks. I also learned how to work on different areas of the project and do a bit of everything.

EMILIO

During the making of this project I learned a lot about teamwork and problem solving. I also improved my skills in databases, which wasn't my strength at first, but I managed to understand it better and apply it throughout the project.



THANK
YOU!

